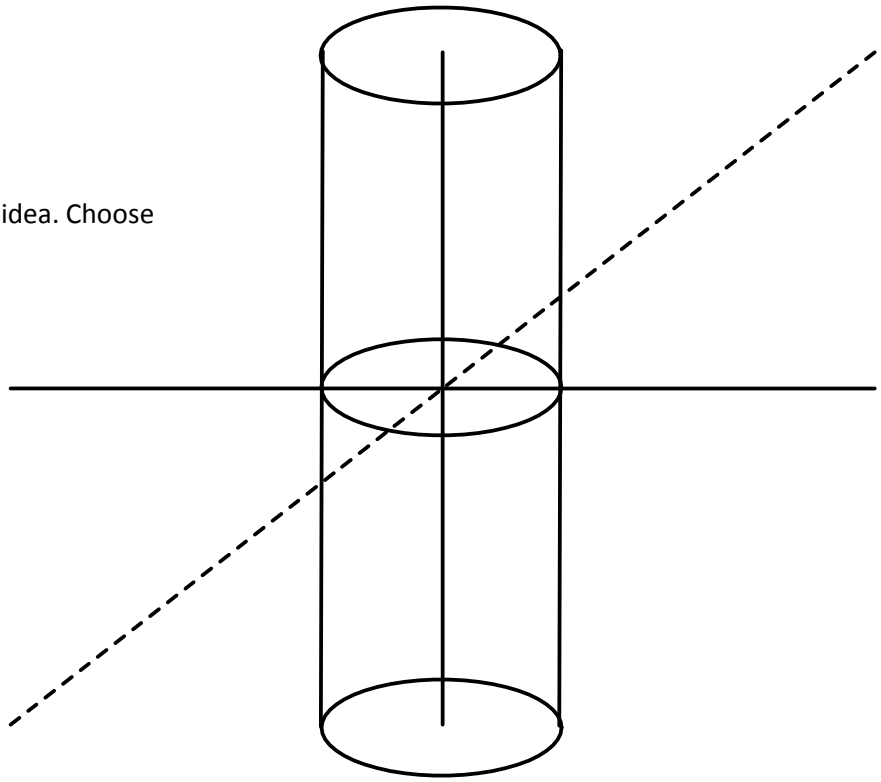
Sec 10.5 Parametric Surfaces

So far we have learned how to parameterize curves but can we do the same thing with a surface? Let's give it a shot. We know that from chapter 9 the equation of a cylinder of radius 2 is $x^2 + y^2 = 4$

How can we write this as a vector equation?



Let's first parameterize some level sets to get the idea. Choose

z=2,4,6

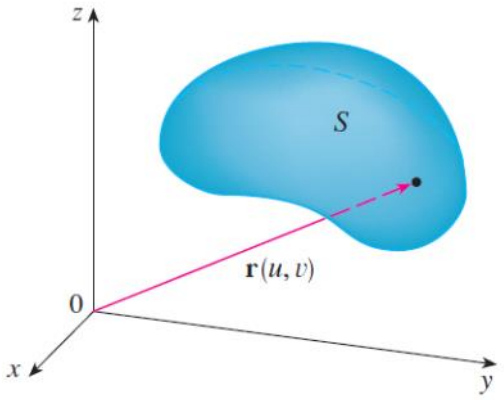
So if we add a parameter we can have our vector equation for the cylinder!

Sec 10.5 Parametric Surfaces

So in general the Vector and Parametric equations of surfaces are of the form.

1 $$\mathbf{r}(u, v) = x(u, v)\,\mathbf{i} + y(u, v)\,\mathbf{j} + z(u, v)\,\mathbf{k}$$

2 $$x = x(u, v) \qquad y = y(u, v) \qquad z = z(u, v)$$

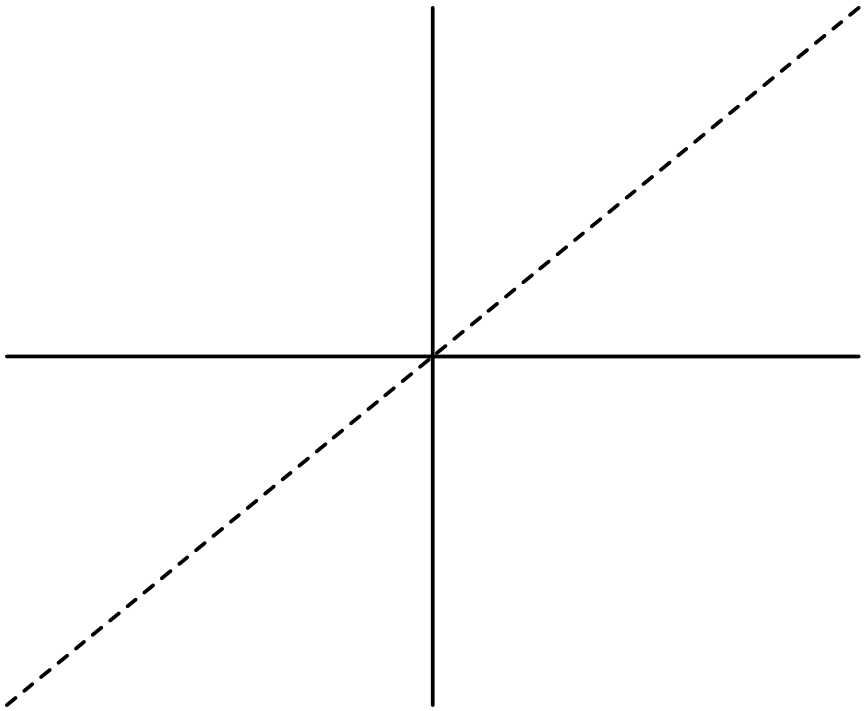


HW example

3–6 Identify the surface with the given vector equation.

3. $\mathbf{r}(u, v) = (u + v)\,\mathbf{i} + (3 - v)\,\mathbf{j} + (1 + 4u + 5v)\,\mathbf{k}$

To understand how to do this we can fix parameter values see what curves we get then see how varying the other parameter moves those curves to trace out a surface. Let's fix some values for u…



Sec 10.5 Parametric Surfaces

Parametric Equation of a Plane: We can define the parametric equation of a plane in general if we have two vectors which are not parallel and which lie in the plane.

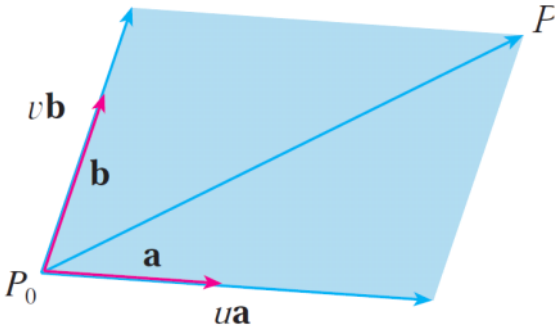


FIGURE 6

Parametric equation of a plane

$$x = x_0 + ua_1 + vb_1 \qquad y = y_0 + ua_2 + vb_2 \qquad z = z_0 + ua_3 + vb_3$$

Where (x_0, y_0, z_0) is a point in the plane and vectors $\vec{a} =< a_1, a_2, a_3 >, \vec{b} =< b_1, b_2, b_3 >$ are any two non parallel vectors which lie in the plane.

Sec 10.5 Parametric Surfaces

HW Example

Find the parametric equations for the surface defined by

23. The part of the sphere $x^2 + y^2 + z^2 = 4$ that lies above the cone $z = \sqrt{x^2 + y^2}$

Here we can simply parameterize the sphere
Then restrict u,v so that we only get points above the cone.